

ARYAN

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SUMMARY

An inquisitive, interdisciplinary researcher majoring in Electronics Engineering with a profound curiosity about the fundamental mechanisms of the human brain. My primary research focus lies at the intersection of genomics and stem cell biology, exploring how the brain develops and functions at a cellular level, complemented by a strong background in electrophysiology and signal analysis to explore neural dynamics. Concurrently, I possess a robust technical foundation in System-on-Chip (SoC) design, hardware-software co-design, embedded systems, FPGA architecture, custom IP development, and robotics. I aim to secure a professional role where I can apply technological and quantitative solutions to frontier challenges in both stem cell research and advanced hardware design.

EDUCATION

University School of Information, Communication and Technology (USICT), GGSIPU Delhi, India
Bachelor of Technology in Electronics and Communication Engineering [GPA: 3.6/4.00] 08/2023 – 05/2027

- **Relevant Coursework & Skills:** Probability & Statistics, University-level Physics (1 & 2), Chemistry (1 & 2), Mathematics (1 & 2), Signals & Systems, Digital Electronics, Analog Electronics, PCB Design, Verilog, VHDL, MATLAB, Simulink, and Embedded Systems.

WORK EXPERIENCE

Bharat Electronics Limited (BEL)   Ghaziabad, India
Intern – SCCS (Satellite and Cellular Communication Systems – Defense) 05/2026 – Present

- Analyzed Zynq-7020 (ZedBoard) architecture, developing custom VHDL logic for Programmable Logic (PL) integrated with bare-metal C programs for Processing System (PS) control via AMBA AXI protocols.
- Engineered a real-time audio spectrogram by capturing live microphone input via the ZedBoard's audio codec using an I2S receiver, transmitting data over UART for dynamic visualization on a PC using PySerial and Matplotlib.
- Assisted in the mechanical assembly, hardware-software integration, and precise tuning of defense-grade Satcom-On-The-Move (SOTM) terminals to validate system performance.

Molecular Biology and Genetics Unit, JNCASR   Bengaluru, India
Research Fellow (SRFP) – Supervisor: Prof. (Dr.) Anuranjan Anand 06/2025 – 09/2025

- Conducted a project titled "Minigene assay to analyze the splicing effect in patients with Orofacial Digital Syndrome and its possible link to epilepsy."
- Cloned genes of interest into plasmids, prepared RNA samples, and analyzed protein interactions and expression patterns.
- Conducted computational docking and RNA-seq data analysis for differential gene expression studies.

Department of Biophysics, AIIMS Delhi   New Delhi, India
Research Intern – Supervisor: Dr. Jyotirmoy Banerjee 01/2025 – 02/2025

- Assisted in analyzing GABAergic and glutamatergic synaptic activity using patch-clamp techniques.
- Processed epileptogenic brain tissue from surgical resection for electrophysiological analysis.
- Handled animal care and mating procedures, alongside data acquisition from single-cell recordings.

AINN – Computation & Neurosurgery   Noida, India
Research Intern – Supervisor: Dr. Birendra Nath Mallick 06/2024 – 08/2024

- Conducted interdisciplinary research combining computational neuroscience and experimental neurosurgery.
- Analyzed rat EEG signals to evaluate the effects of GABAergic modulators across sleep and wakefulness stages.
- Performed stereotaxic surgeries and extracted key time- and frequency-domain EEG metrics using MATLAB.

SKILLS

Programming & Hardware: C, C++, Python, R, MATLAB, Embedded-C, VHDL, Verilog, SystemVerilog, AMBA AXI, RTL Verification.

Tools & Systems: AMD Vivado, Vitis, KiCad, Cadence, Fusion360, Simulink, Linux/Unix, FreeRTOS, ROS2, Git.

Neuroscience & Wet Lab: Animal handling, stereotaxic surgery, electrophysiology (patch-clamp), PCR, SDS-PAGE, Western blotting, RNA extraction, gene cloning, bacterial culture.

AI, Data & Bioinformatics: Signal processing, EEG/sEEG analysis, PyTorch, TensorFlow, OpenCV, NumPy, SciPy, NGS pipelines, scRNA-seq, GWAS analysis, structural docking, PyMOL.

PROJECTS

Custom FPGA Architecture Design



- Designed a custom FPGA architecture from scratch in Cadence, building programmable logic blocks using both standard 6T SRAM and 1T1R memory cells for LUTs.
- Integrated multiplexer (mux) trees and configurable I/O interfaces to ensure reliable data flow and logical connectivity across the board.

Autonomous Omnidirectional Robots for Automated Material Handling



- Engineered custom PCBs with ESP32 microcontrollers to drive three-wheeled omnidirectional robots, tracked in real-time using an OpenCV and ArUco marker-based overhead camera system.
- Implemented Cell Decomposition mapping and the A* search algorithm to compute dynamic routes and guarantee collision avoidance for all three robots simultaneously during crate manipulation.

AXI-Lite Protocol Verification



- Developed a comprehensive functional verification testbench for the AXI-Lite protocol utilizing SystemVerilog and UVM.

Independent Computational Project: Genomic Data Optimization



- Investigated open-source Hi-C datasets and HDF5 data structures to identify computational bottlenecks, conceptualizing a novel data format using information theory to optimize genomic storage and retrieval.

ACHIEVEMENTS & PUBLICATIONS

Research Publication — ICAAI-2025



- Research paper published at ICAAI-2025 titled "*A computational experiment for classification of mesocortical dopaminergic pathways based on human addiction patterns*".

SRFP-2025 Fellowship — JNCASR



Bengaluru, India

- Awarded the highly selective national Summer Research Fellowship Programme (SRFP) to conduct interdisciplinary biomedical research at the Molecular Biology & Genetics Unit.